T2490 — The Discrete-Series Spectrum Theorem: the four dynamical substrate primaries {N_c, n_C, C_2, g} are EXACTLY the half-Casimirs of the four lowest non-trivial holomorphic discrete-series representations of $SO_0(5,2)$, in increasing order. Equivalently, the glueball linear-energy ladder on $H^2(D\_IV^5)$ starts at the genus λ_0 = n_C and steps by spin/parity. An Integer-Web over-determination: the integers that DEFINE the substrate reappear as its spectral levels. Connects the five integers to rep theory directly; underlies the BST glueball spectrum (Lyra F292 linear-energy mechanism).

Grace

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T2490 — The Discrete-Series Spectrum Theorem

Statement (exact)

Let $SO_0(5,2)$ (complexified maximal compact-dual algebra so(7) = B_3) have $\rho = (5/2, 3/2, 1/2)$, $|\rho|^2 = 35/4$. The holomorphic discrete-series representations carry Harish-Chandra parameter $\lambda = \rho + (n_1,n_2,n_3)$ with n_i $\in \mathbb{Z}\geq 0$ and $\lambda_1 > \lambda_2 > \lambda_3 > 0$, and quadratic Casimir $C_2(\lambda) = |\lambda|^2 - |\rho|^2$. Then the four lowest non-trivial Casimir values are 6, 10, 12, 14, and

> ½ · C_2 ∈ {3, 5, 6, 7} = {N_c, n_C, C_2, g} — the four dynamical substrate primaries, in increasing order.

rep λ	n	C_2	½C_2	primary
(3.5, 1.5, 0.5)	(1,0,0)	6	3	N_c
(3.5, 2.5, 0.5)	(1,1,0)	10	5	n_C
(3.5, 2.5, 1.5)	(1,1,1)	12	6	C_2
(4.5, 1.5, 0.5)	(2,0,0)	14	7	g

(rank = 2 is the *rank* of the group itself; N_max = 137 is the Shilov-boundary integer — the two non-spectral primaries. The four *dynamical* primaries are exactly the spectral ones.)

Why it is content, not tautology

The primaries enter the substrate by their *defining* roles — N_c = color/multiplicity = rank^2−1, n_C = dim_$\mathbb{C}$ = genus, C_2 = the (1,1) K-type Casimir, g = signature/embedding. T2490 says these same integers reappear as the lowest discrete-series Casimirs — a different invariant (the discrete-series C_2, not the K-type one). That the defining integers and the spectral levels coincide is an over-determination (Casey's Integer-Web Principle at the spectral level), not a relabeling.

The physical application — the BST glueball spectrum (Lyra F292 mechanism)

The glueball masses are the LINEAR conformal energies of these reps (the dilatation/SO(2) eigenvalue — "remember linear algebra"; *not* the quadratic Casimir, which is the $m^2 \propto C_2$ reading that missed 2^{++}):

m(channel) ∝ E = λ_0 + step, λ_0 = n_C = 5 (the genus / Bergman lowest weight).

channel	step	E	$E/E(0^{++})$	substrate form	lattice	dev
0^{++}	0	5	1	1	1.000	—
2^{++}	spin-2 = rank	7	7/5	g/n_C (BLIND)	1.387	0.9%
0^{-+}	twist n_C/2	7.5	3/2	N_c/rank	1.497	0.2%
1^{+-}	spin-1 + twist	8.5	17/10	17/10	1.699	0.04%

The blind leg (the derivation): $2^{++}/0^{++}$ = (n_C + rank)/n_C = g/n_C, using only the genus + spin-2 step and the substrate identity g = n_C + rank (7 = 5 + 2) — nothing read from the data. This is the leg that beats look-elsewhere: the spectrum collapses to two BST-natural numbers (λ_0 = n_C, twist = n_C/2), not four free ratios. (Honest tier: 2^{++} blind; $0^{-+}/1^{+-}$ use the n_C/2 half-canonical twist, rep-motivated but value-checked against data — pending Elie's two confirming toys. So T2490's *spectral ladder* is D-tier exact; the *glueball mass-map* is I-tier with the 2^{++} leg blind.)

Connections (graph edges)

- Casey's Integer-Web Principle — T2490 is a spectral instance: defining integers = spectral levels.
- g = n_C + rank (7 = 5 + 2) and C_2 = n_C + 1 — the low ladder rungs are primary-sums anchored at the genus n_C. (Edges to the primary-identity nodes.)
- T1829 (Wallach Bottleneck) — the K-type/Casimir machinery of the same discrete series; T2490 is its conformal-energy companion (Casimir ladder ↔ linear-energy ladder, same reps).
- Compact tower k(k+5) = {0,6,14,24} and noncompact k(k+4) = {0,5,12,21} (corpus) — share rungs 6 = C_2 and 14 = 2g with the T2490 ladder {0,6,10,12,14}: a shared spectral spine.
- Paper A Yang-Mills gap Δ = C_2 = 6 — the first Casimir rung of the ladder; the mass gap is the ladder's first step. (Edge: T2490 → the YM gap value.)
- Lyra F292 (linear-energy mechanism) and T2489 (HS mirror; these are holomorphic discrete-series objects, Hardy-paired → HS-mirror nodes exist).

AC and tier

AC = (C=1, D=1), depth 0 for the spectral-ladder statement (a finite Casimir computation in the discrete-series Weyl chamber — counting + identity). The glueball mass-map is the physical application (I-tier, 2^{++} blind).

— T2490, registered by Grace, 2026-06-23. Spectral connection mine; glueball linear-energy mechanism Lyra (F292); forced addresses Elie. For Keeper registry + graph insertion; for Cal cold-read.